Estimating the prevalence of the Sonority Sequencing Principle accross languages and language families

Method

Dataset acquisition

All 8050 languages included in the Glottolog database were targeted. Using the lingtypology Python package (Voronov and Forkel 2019; Moroz 2017), we extracted their corresponding ISO-639-3 code and accompanying information (e.g., language family, official name). We then used the wikipron Python package (Lee et al. 2020) to scrape the Wiktionary page of each language looking for pronunciations. First, narrow phonological transcriptions were searched. If no narrow phonological transcriptions are available, broad phonological transcriptions were searched. Finally, if no broad phonological transcriptions are available either, the language was skipped. We then filtered our languages with less than 100 available pronunciations. This resulted in 247 languages with at least 100 pronunciations available in Wiktionary.

For each language, and for each retrieved pronunciation, we chunked the phonological word-forms into syllable candidates by splitting the IPA transcription at syllabic boundaries, primary and secondary stress markers. The dataset of each language was then expanded to assign one row to each syllable candidate. Languages with less than 20 unique syllables were filtered out. We then used the panphon Python package (Mortensen et al. 2016) to extract the features of each syllable candidate. First, we split the phonological form of the syllable candidate into phonological segments. Second, we classified each segment as consonant (C) or vowel (V). Glides were classified as vowels. Third, we counted the number of vocalic clusters (number of group of consecutive segments classified as vocalic) in each syllable candidate. If more than cluster was detected, the syllable candidate was considered to include more than one syllable and was filtered out. This was the case when the original phonological form of the word retrieved from Wiktionary did not include syllabic boundaries. Finally, we calculated the sonority contour of each syllable: using panphon, we extracted the sonority score of each segment (an integer ranging from 1 to 9). If the sonority contour had more than one local

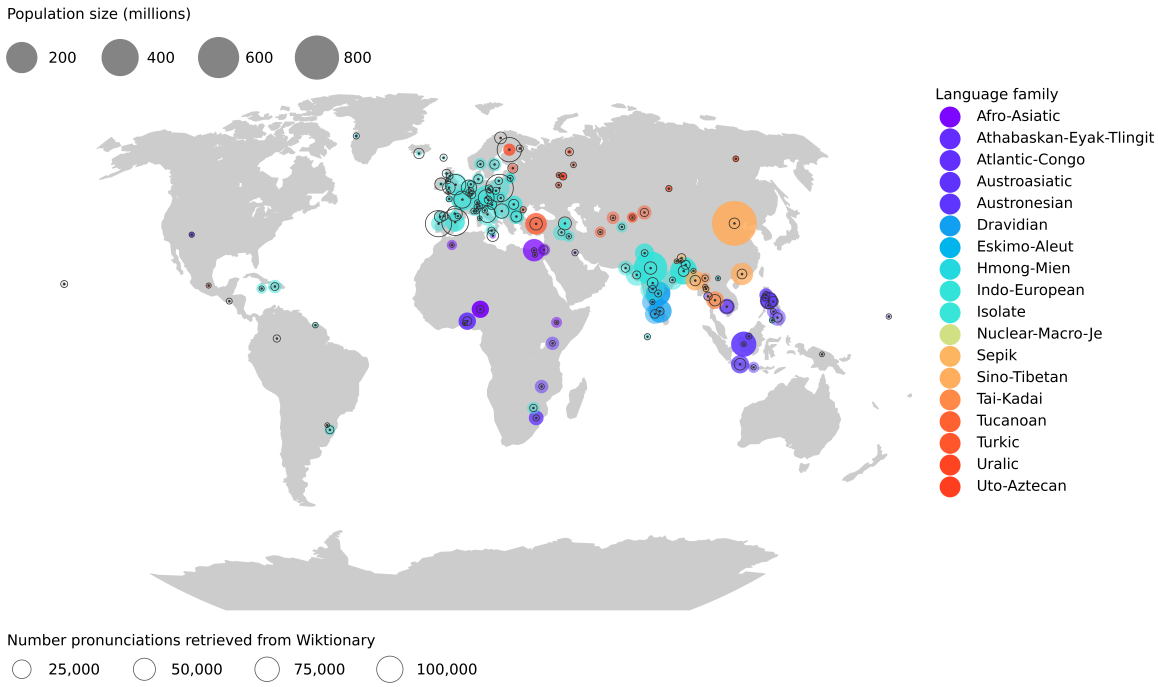


Figure 1: Distribution of the 247 languages included in the final dataset. Black dots indicate the location assigned to each language by Glottolog. The color and size of the shaded circles indicate the language family and estimated population of speakers of each language, respectively. The size of the empty black circles indicate the number of pronunciations retrived from each language. Large divergencies between the size of the shaded and empty circles indicate the the language is over or under-represented in the dataset.

maximum, the syllable was classified as violating the Sonority Sequencing Principle (SSP). If the sonority contour had no peaks (flat) or had only one peak, the syllable was classified as following the SSP.

Data analysis

We used a multilevel, Bayesian logistic regression model to estimate the proportion of syllables that followed or violated the SSP across languages. First, we fit a model that estimated the average probability of the syllables of each language following the SSP. This model included each language as a group with a random intercept. Second, we fit a model that estimated the average probability of the syllables of language family following the SSP. This model included each language as a group, also with a random intercept. Equation 1 shows a detailed implementation of both models model. We implemented the model in Python 3.12.7 using the PyMC package (Abril-Pla et al. 2023), with four Monte Carlo chains, each with 2,000 iterations. Model diagnostics and posterior distribution preprocessing were conducted using the ArViZ Python package (Kumar et al. 2019).

Likelihood:

$$
\begin{aligned}
y_{[j]i} &\sim \text{Binomial}(n_i, \frac{\theta_{[j]i}}{1-\theta_{[j]i}}) \\
\theta_{j[i]} &\sim \mathcal{N}(\mu_j, \sigma_j) \\
&\text{where:} \\
&\quad i \text{ represents each observation, corresponding to a language} \\
&\quad j \text{ represents each language family} \\
&\quad y_{[j]i} \text{ is the number of syllables in language } i \text{ that follow SSP} \\
&\quad n_j \text{ is the number of syllables total language } i \\
\textbf{Prior:} \mu_j &\sim \mathcal{N}(0, 1) \\
\sigma_j &\sim \text{Exponential}(4)
\end{aligned}
\tag{1}
$$

Results

Estimating the prevalence of the SSP across languages

Appendix

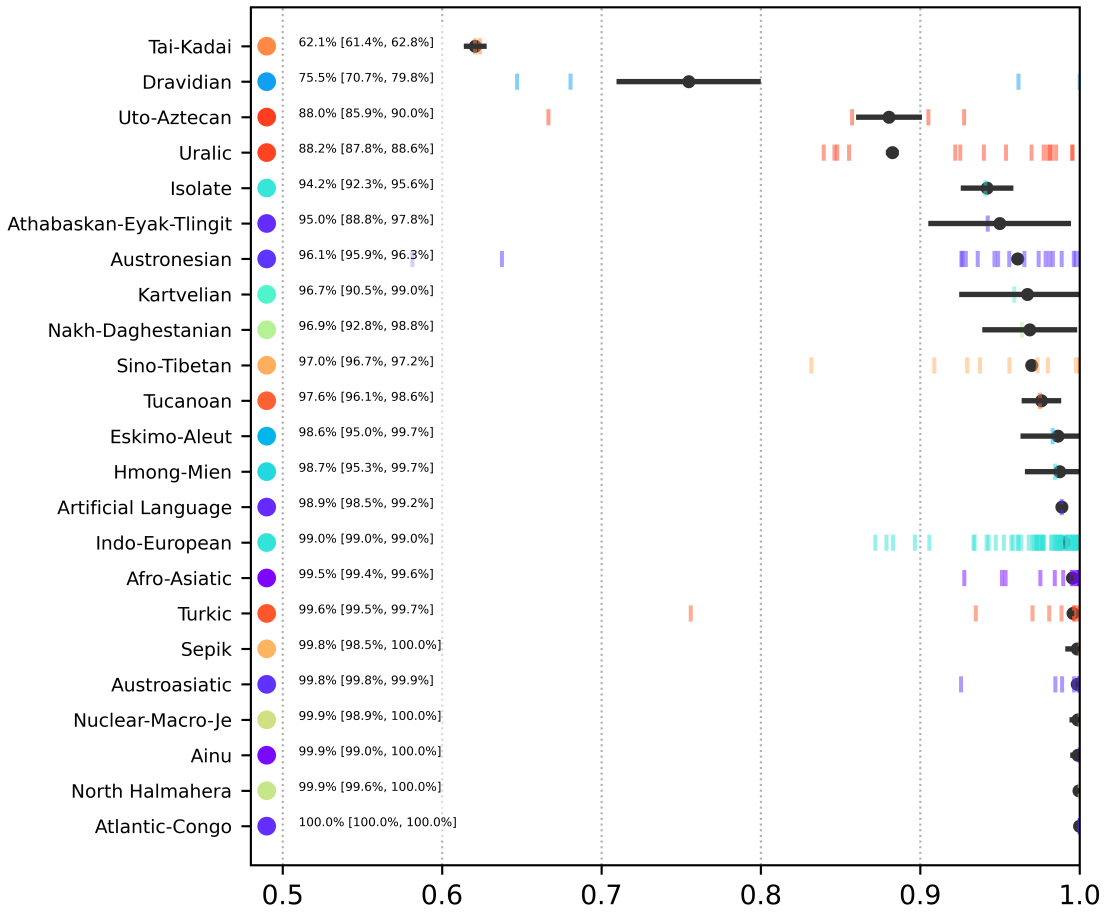


Figure 2: Posterior model-estimated proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language family. The X-axis indicates the proportion of syllables that follow the SSP. The Y-axis indicates each language family. Black diamonds and intervals indicate the posterior mean proportion of SSP-following syllables for each language family, and its 89% highest density interval (HDI). Means and 89% HDI are also numerically reported next to each language family.

Table 1: Observed proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language.

Language	Language family	# Words	# Syllables	SSP (%)
Hijazi Arabic	Afro-Asiatic	952	1,901	99.58%
Hausa	Afro-Asiatic	738	6,928	100.00%
Afar	Afro-Asiatic	1,335	1,559	99.81%
Sidamo	Afro-Asiatic	101	43	95.35%
Akkadian	Afro-Asiatic	184	454	100.00%
Gulf Arabic	Afro-Asiatic	343	447	98.43%
Modern Hebrew	Afro-Asiatic	948	1,642	98.96%
Moroccan Arabic	Afro-Asiatic	833	1,371	99.78%
Maltese	Afro-Asiatic	3,777	9,384	99.49%
Egyptian (Ancient)	Afro-Asiatic	942	469	92.75%
Libyan Arabic	Afro-Asiatic	164	81	97.53%
Central Moroccan Berber	Afro-Asiatic	108	41	95.12%
Egyptian Arabic	Afro-Asiatic	148	398	100.00%
Coptic	Afro-Asiatic	128	145	100.00%
South Levantine Arabic	Afro-Asiatic	1,186	2,387	99.96%
Hokkaido Ainu	Ainu	130	304	100.00%
Esperanto	Artificial Language	5,975	3,089	98.87%
Navajo	Athabaskan-Eyak-Tlingit	168	104	94.23%
Nyanja	Atlantic-Congo	197	498	100.00%
Kikuyu	Atlantic-Congo	207	54	100.00%
Nupe-Nupe-Tako	Atlantic-Congo	153	277	100.00%
Gun	Atlantic-Congo	143	293	100.00%
Zulu	Atlantic-Congo	329	40	100.00%
Yoruba	Atlantic-Congo	1,311	9,261	100.00%
Central Khmer	Austroasiatic	6,070	24,306	99.92%
Eastern Lawa	Austroasiatic	254	272	99.63%
Bahnar	Austroasiatic	198	131	98.47%
Pnar	Austroasiatic	102	94	92.55%
Muong	Austroasiatic	195	195	100.00%
Mon	Austroasiatic	1,051	1,077	98.89%
Nyahkur	Austroasiatic	152	187	100.00%
Tagalog	Austronesian	13,663	37,929	95.57%
Iloko	Austronesian	510	976	94.88%
Betawi	Austronesian	153	42	92.86%
Kankanaey	Austronesian	422	375	96.53%
Brunei	Austronesian	159	140	97.86%
Tausug	Austronesian	299	43	58.14%
Hawaiian	Austronesian	503	2,732	100.00%

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Language	Language family	# Words	# Syllables	SSP (%)
Cebuano	Austronesian	1,556	1,265	93.60%
Hiligaynon	Austronesian	230	202	92.57%
Marshallese	Austronesian	185	80	63.75%
Pampanga	Austronesian	313	368	98.10%
Indonesian	Austronesian	3,851	7,756	98.87%
Balinese	Austronesian	164	194	97.42%
Coastal-Naga Bikol	Austronesian	2,165	2,227	92.64%
Javanese	Austronesian	145	798	99.62%
Kawi	Austronesian	218	396	99.75%
Iban	Austronesian	254	178	98.31%
Hanunoo	Austronesian	239	243	94.65%
Malayalam	Dravidian	1,166	72	68.06%
Kannada	Dravidian	146	26	100.00%
Tamil	Dravidian	887	170	64.71%
Telugu	Dravidian	527	78	96.15%
Kalaallisut	Eskimo-Aleut	289	118	98.31%
Hmong Daw	Hmong-Mien	124	130	98.46%
Kashmiri	Indo-European	136	231	96.10%
Gujarati	Indo-European	870	1,890	99.89%
Irish	Indo-European	4,682	4,512	97.70%
Sassarese Sardinian	Indo-European	108	74	100.00%
Sranan Tongo	Indo-European	210	130	100.00%
Occitan	Indo-European	243	344	99.71%
Spanish	Indo-European	106,961	107,264	99.74%
Lower Sorbian	Indo-European	647	479	97.29%
Eastern Armenian	Indo-European	5,906	15,060	97.58%
Emiliano	Indo-European	167	169	97.63%
Ems-Weser Frisian	Indo-European	163	112	97.32%
Scottish Gaelic	Indo-European	1,112	667	98.80%
Bengali	Indo-European	2,771	5,097	97.59%
Galician	Indo-European	3,477	2,668	99.93%
Haitian	Indo-European	848	513	98.83%
Welsh	Indo-European	5,150	2,834	97.49%
Jamaican Creole English	Indo-European	150	129	98.45%
Central Alemannic	Indo-European	159	105	94.29%
Piemontese	Indo-European	264	285	98.60%
Old French (842-ca. 1400)	Indo-European	295	344	97.09%
Eastern Panjabi	Indo-European	452	1,348	98.81%

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Language	Language family	# Words	# Syllables	SSP (%)
Middle English	Indo-European	2,827	1,376	95.71%
Corsican	Indo-European	110	83	100.00%
Avestan	Indo-European	121	64	98.44%
Walloon	Indo-European	612	1,116	99.64%
Scots	Indo-European	370	340	97.35%
Faroese	Indo-European	563	351	87.18%
Swedish	Indo-European	1,624	1,526	99.21%
Old Spanish	Indo-European	165	133	98.50%
Modern Greek	Indo-European	3,104	19,664	100.00%
Upper Sorbian	Indo-European	372	180	93.33%
Old English (ca. 450-1100)	Indo-European	12,302	33,329	98.88%
Central Kurdish	Indo-European	184	66	87.88%
Kashubian	Indo-European	689	1,493	98.66%
Danish	Indo-European	1,399	626	97.28%
Italian	Indo-European	15,211	54,183	99.96%
Cornish	Indo-European	133	86	98.84%
Lombard	Indo-European	199	209	98.56%
Assamese	Indo-European	1,379	2,644	99.70%
Serbian-Croatian-Bosnian	Indo-European	8,822	870	89.66%
Sicilian	Indo-European	519	977	100.00%
French	Indo-European	17,659	98,238	99.78%
Chavacano	Indo-European	162	124	100.00%
Polish	Indo-European	123,564	386,285	98.19%
Dutch	Indo-European	11,340	27,741	98.74%
Hindi	Indo-European	5,072	40,764	99.85%
Lithuanian	Indo-European	279	874	99.77%
Manx	Indo-European	186	147	99.32%
Early Irish	Indo-European	392	284	100.00%
Hunsrik	Indo-European	888	599	96.33%
Dhivehi	Indo-European	326	705	99.57%
Aragonese	Indo-European	178	146	100.00%
Tajik	Indo-European	258	162	98.77%
Fala	Indo-European	183	140	100.00%
Romanian	Indo-European	2,126	9,710	99.53%
Ancient Greek	Indo-European	28,165	100,339	99.99%
Northern Kurdish	Indo-European	508	728	99.45%
Breton	Indo-European	286	760	99.21%
Slovak	Indo-European	487	114	94.74%

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Language	Language family	# Words	# Syllables	SSP (%)
Phalura	Indo-European	365	91	93.41%
Limburgan	Indo-European	205	356	99.16%
Moselle Franconian	Indo-European	860	693	95.24%
German	Indo-European	13,142	21,590	90.56%
Asturian-Leonese-Cantabrian	Indo-European	1,181	901	100.00%
Bavarian	Indo-European	756	649	88.29%
Urdu	Indo-European	1,088	2,199	99.73%
Western Frisian	Indo-European	425	314	97.77%
Odia	Indo-European	258	26	96.15%
Icelandic	Indo-European	2,346	1,180	94.15%
Gothic	Indo-European	404	456	95.83%
Ligurian	Indo-European	194	186	99.46%
English	Indo-European	36,929	48,590	96.80%
Norwegian	Indo-European	451	305	97.70%
Portuguese	Indo-European	99,468	361,689	99.94%
Sanskrit	Indo-European	2,831	7,453	99.80%
Slovenian	Indo-European	1,704	440	97.27%
Old Russian	Indo-European	233	168	97.02%
Silesian	Indo-European	518	1,087	98.25%
Afrikaans	Indo-European	619	970	99.07%
Marathi	Indo-European	703	1,586	100.00%
Basque	Isolate	10,555	778	94.09%
Laz	Kartvelian	143	73	95.89%
Bats	Nakh-Daghestanian	253	138	96.38%
Ternate	North Halmahera	122	273	100.00%
West Makian	North Halmahera	325	725	100.00%
Kaingang	Nuclear-Macro-Je	107	300	100.00%
Abau	Sepik	119	179	100.00%
Yamphu	Sino-Tibetan	137	32	93.75%
Tibetan	Sino-Tibetan	1,208	1,907	83.17%
Mandarin Chinese	Sino-Tibetan	2,577	7,548	99.76%
Balti	Sino-Tibetan	101	77	100.00%
Dzongkha	Sino-Tibetan	104	136	95.59%
Burmese	Sino-Tibetan	2,430	1,184	90.88%
Khiamniungan Naga	Sino-Tibetan	2,952	650	98.00%
Western Kayah	Sino-Tibetan	129	151	97.35%
Yue Chinese	Sino-Tibetan	1,875	3,740	100.00%
Zou	Sino-Tibetan	145	170	92.94%

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Language	Language family	# Words	# Syllables	SSP (%)
Limbu	Sino-Tibetan	107	190	100.00%
S'gaw Karen	Sino-Tibetan	220	250	100.00%
Shan	Tai-Kadai	361	731	62.38%
Thai	Tai-Kadai	3,564	16,974	62.06%
Cubeo	Tucanoan	582	602	97.51%
Sakha	Turkic	214	822	100.00%
Uighur	Turkic	967	168	75.60%
Turkish	Turkic	5,108	22,552	99.76%
Salar	Turkic	325	52	98.08%
Kirghiz	Turkic	181	548	99.64%
Kazakh	Turkic	195	288	99.65%
Tuvinian	Turkic	313	1,146	100.00%
Turkmen	Turkic	157	135	97.04%
Gagauz	Turkic	311	438	98.86%
Turkic Khalaj	Turkic	139	46	93.48%
North Saami	Uralic	4,114	128	92.19%
Liv	Uralic	387	99	96.97%
Votic	Uralic	1,876	202	98.51%
Erzya	Uralic	236	52	84.62%
Western Mari	Uralic	139	57	98.25%
Kildin Saami	Uralic	659	366	93.99%
Komi-Permyak	Uralic	229	224	99.55%
Livvi	Uralic	284	40	92.50%
Finnish	Uralic	73,343	16,004	84.78%
Moksha	Uralic	334	396	97.73%
Komi-Zyrian	Uralic	908	964	98.13%
Karelian	Uralic	501	76	85.53%
Ingrian	Uralic	8,558	1,114	83.93%
Eastern Mari	Uralic	659	2,481	99.52%
Akkala Saami	Uralic	180	108	95.37%
Estonian	Uralic	2,050	991	97.98%
Classical Nahuatl	Uto-Aztecan	865	537	90.50%
Isthmus-Mecayapan Nahuatl	Uto-Aztecan	147	75	66.67%
Tlaxcala-Puebla-Central Nahuatl	Uto-Aztecan	146	124	92.74%
Pipil	Uto-Aztecan	269	175	85.71%

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